

Assignment #10: dp & bfs

Updated 2 GMT+8 Nov 25, 2024

2024 fall, Compiled by 同学的姓名、院系

说明:

1) 请把每个题目解题思路 (可选), 源码 Python, 或者 C++ (已经在 Codeforces/Openjudge 上 AC), 截

图 (包含 Accepted), 填写到下面作业模版中 (推荐使用 typora <https://typoraio.cn>, 或者用 word)。AC 或者没有 AC, 都请标上每个题目大致花费时间。

2) 提交时候先提交 pdf 文件, 再把 md 或者 doc " " 文件上传到右侧 作业评论。Canvas 需要有同学清晰头像、提

交文件有 pdf、"作业评论"区有上传的 md 或者 doc 附件。

3) 如果不能在截止前提交作业, 请写明原因。

1. 题目

LuoguP1255 数楼梯

dp, bfs, <https://www.luogu.com.cn/problem/P1255>

思路:

代码:

```
dp = [0]*5001
dp[1] = 1
dp[2] = 2
for i in range(3,5001):
    dp[i] = dp[i-1] + dp[i-2]
m = int(input())
print(dp[m])
```

代码运行截图 (至少包含有"Accepted")

源代码 [复制](#)

```
dp = [0]*5001
dp[1] = 1
dp[2] = 2
for i in range(3,5001):
    dp[i] = dp[i-1] + dp[i-2]
m = int(input())
print(dp[m])
```

所属题目

P1255 数楼梯

评测状态

Accepted

提交时间

2024-11-27 21:10:35

27528: 跳台阶

dp, <http://cs101.openjudge.cn/practice/27528/>

思路:

代码:

```

dp = [0]*26
dp[0] = 1
dp[1] = 1
for i in range(2,26):
    for j in range(0,i):
        dp[i] +=dp[j]
n = int(input())
print(dp[n])

```

代码运行截图 (至少包含有"Accepted")

状态: Accepted

源代码

```

dp = [0]*26
dp[0] = 1
dp[1] = 1
for i in range(2,26):
    for j in range(0,i):
        dp[i] +=dp[j]
n = int(input())
print(dp[n])

```

基本信息

#: 47434258
 题目: 27528
 提交人: 24n2400011491
 内存: 3640kB
 时间: 34ms
 语言: Python3
 提交时间: 2024-11-27 21:35:09

474D. Flowers

dp, <https://codeforces.com/problemset/problem/474/D>

思路:

代码:

```

t,k = map(int,input().split())

dp = [0]*100001
dp[0] = 1
s = [0]*100001
for i in range(1,100001):
    if i >=k:
        dp[i] = (dp[i-1] + dp[i-k])%(10**9+7)
    else:
        dp[i] = dp[i-1]
    s[i] = (s[i-1] +dp[i])%(10**9+7)
for _ in range(t):
    a,b = map(int,input().split())
    print((s[b]-s[a-1])%(10**9+7))

```

代码运行截图 (至少包含有"Accepted")

```
t, k = map(int, input().split())

dp = [0]*100001
dp[0] = 1
s = [0]*100001
for i in range(1, 100001):
    if i >= k:
        dp[i] = (dp[i-1] + dp[i-k])%(10**9+7)
    else:
        dp[i] = dp[i-1]
    s[i] = (s[i-1] + dp[i])%(10**9+7)
for _ in range(t):
    a, b = map(int, input().split())
    print((s[b]-s[a-1])%(10**9+7))
```

LeetCode5.最长回文子串

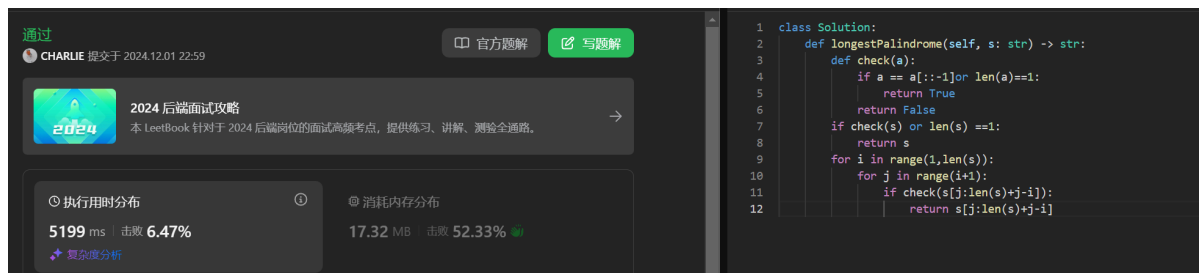
dp, two pointers, string, <https://leetcode.cn/problems/longest-palindromic-substring/>

思路：从整个字符串的长度开始，不断筛选长度越来越小的回文串

代码：

```
class Solution:
    def longestPalindrome(self, s: str) -> str:
        def check(a):
            if a == a[::-1] or len(a)==1:
                return True
            return False
        if check(s) or len(s) ==1:
            return s
        for i in range(1, len(s)):
            for j in range(i+1):
                if check(s[j:len(s)+j-i]):
                    return s[j:len(s)+j-i]
```

代码运行截图 (至少包含有"Accepted")



12029: 水淹七军

bfs, dfs, <http://cs101.openjudge.cn/practice/12029/>

思路：

代码：

```
import sys
from collections import deque
data = sys.stdin.read().split()

move = [[1,0],[0,1],[-1,0],[0,-1]]
```

```

k = int(data[0])
id = 1
for _ in range(k):
    m,n = int(data[id]),int(data[id+1])
    id +=2
    ditu = [[1001 for _ in range(n+2)] for _ in range(m+2)]
    w_h = [[0 for _ in range(n+2)] for _ in range(m+2)]
    for j in range(1,m+1):
        ditu[j][1:n+1] = list(map(int,data[id:id+n]))
        id +=n
    a,b = int(data[id]),int(data[id+1])
    id +=2
    p = int(data[id])
    id +=1
    def water(q,start):
        while q:
            x,y = q.pop()
            w_h[x][y] = start
            for k in range(4):
                nx = x + move[k][0]
                ny = y + move[k][1]
                if start > ditu[nx][ny] and start > w_h[nx][ny]:
                    q.append([nx, ny])

    for i in range(p):
        q=[(int(data[id]),int(data[id+1]))]
        water(q,ditu[int(data[id])][int(data[id+1])])
        id +=2

    if w_h[a][b] >0:
        print('Yes')
    else:
        print('No')

```

代码运行截图 (至少包含有"Accepted")

状态: Accepted

基本

源代码

```
import sys
from collections import deque
data = sys.stdin.read().split()

move = [[1,0],[0,1],[-1,0],[0,-1]]

k = int(data[0])
id = 1
for _ in range(k):
    m,n = int(data[id]),int(data[id+1])
    id +=2
    ditu = [[1001 for _ in range(n+2)] for _ in range(m+2)]
    w_h = [[0 for _ in range(n+2)] for _ in range(m+2)]
    for j in range(1,m+1):
        ditu[j][1:n+1] = list(map(int,data[id:id+n]))
        id +=n
    a,b = int(data[id]),int(data[id+1])
    id +=2
    p = int(data[id])
    id +=1
    def water(q,start):
        while q:
            x,y = q.pop()
            w_h[x][y] = start
            for k in range(4):
                nx = x + move[k][0]
                ny = y + move[k][1]
                if start > ditu[nx][ny] and start > w_h[nx][ny]:
                    q.append([nx, ny])

    for i in range(p):
        q=[(int(data[id]),int(data[id+1]))]
        water(q,ditu[int(data[id])][int(data[id+1])])
        id +=2

    if w_h[a][b] >0:
        print('Yes')
    else:
        print('No')
```

提

02802: 小游戏

bfs, <http://cs101.openjudge.cn/practice/02802/>

思路:

代码:

```
from collections import deque
import copy

move = [[0,1],[0,-1],[1,0],[-1,0]]
ans = []
def bfs(matrix,x1,y1,x2,y2,visit):
    q = deque([(x1,y1,0,-1)])
    while q:
        x,y,seq,j = q.popleft()
        visit[y][x] = True
```

```

        for i in range(4):
            nx,ny = x+move[i][0],y+move[i][1]
            if matrix[ny][nx] == ' ' and not visit[ny][nx]:
                if j != i:
                    q.append((nx,ny,seq+1,i))
                else:
                    q.append((nx,ny,seq,i))
            elif nx == x2 and ny == y2:
                if j != i:
                    ans.append(seq+1)
                else:
                    ans.append(seq)

m = 0
while True:
    m+=1
    w,h = map(int,input().split())
    if w == 0 and h == 0:
        break
    print(f'Board #{m}:')

    visited = [[False for i in range(w+4)] for j in range(h+4)]
    matrix = [['X' for i in range(w+4)] for j in range(h+4)]
    for i in range(1,h+3):
        matrix[i][1:w+3] = [' ']*(w+2)
    for i in range(2,h+2):
        matrix[i][2:w+2] = input()

    n = 0
    while True:
        n+=1
        x1,y1,x2,y2 = map(int,input().split())
        if x1 == y1 == x2 == y2 == 0:
            break
        ans = []
        v = copy.deepcopy(visited)
        bfs(matrix,x1+1,y1+1,x2+1,y2+1,v)
        if not ans:
            print(f'Pair {n}: impossible.')
        else:
            print(f'Pair {n}: {min(ans)} segments.')
    print()

```

代码运行截图 (至少包含有"Accepted")

状态: Accepted

源代码

```
from collections import deque
import copy

move = [[0,1],[0,-1],[1,0],[-1,0]]
ans = []
def bfs(matrix,x1,y1,x2,y2,visit):
    q = deque([(x1,y1,0,-1)])
    while q:
        x,y,seq,j = q.popleft()
        visit[y][x] = True
        for i in range(4):
            nx,ny = x+move[i][0],y+move[i][1]
            if matrix[ny][nx] == '.' and not visit[ny][nx]:
                if j != i:
                    q.append((nx,ny,seq+1,i))
                else:
                    q.append((nx,ny,seq,i))
            elif nx == x2 and ny == y2:
                if j != i:
                    ans.append(seq+1)
                else:
                    ans.append(seq)
m = 0
while True:
    m += 1
```

基本信息

#: 47518241
题目: 02802
提交人: 24n2400011491
内存: 4900kB
时间: 240ms
语言: Python3
提交时间: 2024-12-02 17:20:40

2. 学习总结和收获

如果作业题目简单，有否额外练习题目，比如：OJ“计概 2024fall”每日选做、CF、LeetCode、洛谷等网站题目。

这次作业的题目感觉思路并不难想，就是因为各种原因出现类似runtime error的小错误，看来也是平时练习少导致的。

最近刚刚感冒，这周末主要就是生病了，我打算下周补一补每日选做，保持手感。

顺带一提，我感觉现在的AI并不擅长给代码挑错，很多时候我遇到一些小问题AI发现不了。